STEVENS INSTITUTE OF TECHNOLOGY

Data Acquisition and Processing I: Big Data

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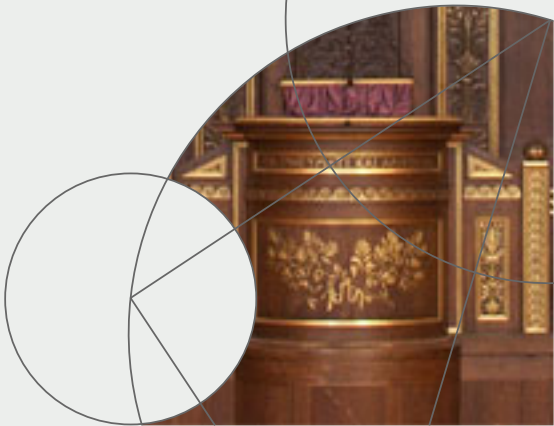
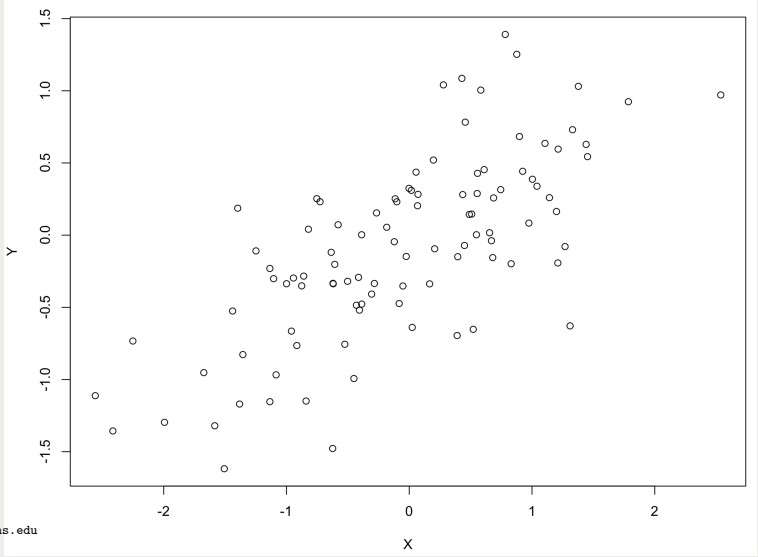


Correlation

- The correlation coefficient, r, is a summary measure that describes the extent of the statistical relationship between two variables X vs. Y.



Correlation Coefficient

- The correlation coefficient between X and Y is defined as

$$r_{x,y} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = \frac{E[(X-\mu_x)(Y-\mu_y)]}{\sqrt{E(X-\mu_x)^2 E(Y-\mu_y)^2}}$$

where

$$\begin{aligned} \mu_x &: \text{ the mean of } X \\ \mu_y &: \text{ the mean of } Y \\ \mathrm{Var}(X) &: \text{ the variance of } X \\ \mathrm{Var}(Y) &: \text{ the variance of } Y \end{aligned}$$

- The correlation factor $r_{x,y}$ means the strength of linear dependence between X and Y.
- The correlation factor $r_{x,y}$ has $-1 \le r_{x,y} \le 1$.
- $r_{x,y} = 0$ means X and Y are uncorrelated (however, uncorrelated $\neq$ independent).

Interpretation of r

- It can be seen there that the possible values range from -1 to $+1$.
- A correlation coefficient close to -1 indicates that the values of X and Y are strongly negatively associated. That is, for larger values of X, it is generally found that the values of Y are smaller. Alternatively stated, if r is close to -1, as X increases, Y generally decreases.
- When r is close to 0, either on the positive or the negative side, then there is little or no association between X and Y .
- When the correlation coefficient is above 0, then this provides evidence of a positive relationship between X and Y . That is, if $r > 0$, larger values of X are associated with larger values of Y .

Why Correlation Coefficients?

- Prediction

  If there is a relationship between two variables, we can make predictions about one from another.

- Validity

  Concurrent validity (correlation between a new measure and an established measure).

- Reliability

  Test-retest reliability (are measures consistent).

  Inter-rater reliability (are observers consistent).

- Theory verification

  Predictive validity.